Fiscal Year:	FY 2015	**Task Last Updated:**	FY 10/06/2015
PI Name:	Quintiere, James Ph.D.		
Project Title:	Experimental Investigation of Emulated Burning Rate at Various Gravity Levels		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jimq@umd.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	301-405-3993
Organization Name:	University of Maryland		
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PI Web Page:			
City:	College Park	**State:**	MD
Zip Code:	20742-3031	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2010	**End Date:**	12/31/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
Contact Email:	david.l.urban@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date extended to 12/31/2014 per NSSC information (Ed., 9/17/13) NOTE: End date extended to 12/31/2013 per PI (Ed., 10/12/2012)		
Key Personnel Changes/Previous PI:	Not applicable.		
COI Name (Institution):	Sunderland, Peter (University of Maryland)		
Grant/Contract No.:	NNX10AD98G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A novel experimental apparatus, the Burning Rate Emulator (BRE), is being developed. This apparatus is capable of assessing ignitability, steady burning rate, and extinction characteristics of 'virtual' solid and liquid fuels as a function of gravity level, flow characteristics, and orientation. By changing the fuel gas and its flow rate, it will be possible to emulate steady combustion for a range of liquid and solid fuels having B numbers (mass transfer numbers) and radiation characteristics that fall anywhere within physically realistic bounds. Measurements of the critical mass flux at ignition can also be made. These measurements will be conducted as a function of scale and orientation to assess turbulent and laminar burning in natural convection. The experiment is well suited to testing in the Combustion Integrated Rack aboard the International Space Station. The Burning Rate Emulator (BRE) will simulate the steady burning of liquids and solids using a rectangular porous burner fed by gaseous fuel and equipped with a total heat flux sensor. BRE will control the fuel flow rate based on the measured heat flux at the burner and the measured surface temperature. This will allow it to use a small number of gaseous fuels to match the following four properties of a large number of condensed fuels: heat of combustion (LHV), heat of vaporization (L), surface temperature (Ts), and laminar smoke point. Extinguishment will be considered via changes in fuel flow rate. Both low flow rates (for kinetic extinction) and high flow rates (for radiative extinction) will be considered. The BRE flight program seeks to establish the burning conditions in a quiescent microgravity ambient in terms of material properties. The specific flight objectives are as follows. 1. Observe burning behavior in long-duration microgravity for CH4/N2 and C2H4/N2 mixtures on porous circular burners of 25 and 50 mm diameter. Consider O2/N2 ambients with various pressures and ambient O2 concentrations and pressures. NASA-considered atmosphere for human space missions will be examined: 14.7 psia, 21% oxygen: 10.2 psia, 26.5%; and 8.2 psia, 34% oxygen. Emphasize fuel conditions that can be related to a diverse range of condensed fuels. 2. Consider the conditions of Objective 1 for ambients of O2/CO2 (desired). 3. Consider the conditions of Objectives 1 – 2 for extinction limits. NOTE: This project is continued as a new grant with the same Principal Investigator--"Experimental Investigation of Emulated Burning Rate at Various Gravity Levels--NNX15AD06A." See that task for subsequent reporting. (Ed., 10/1/2015).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is expected to improve the understanding of fire and combustion processes in both microgravity and normal gravity. It also could advance the state of the art in the design of porous burners and simple heat flux meters.
Task Progress:	[Ed. Note (10/6/15): Compiled from Final Report submitted December 2014] • One M.S. thesis was defended and submitted in Dec., 2014. • One journal paper was published, and another submitted in Dec., 2014. • A poster was presented at the Combustion Symposium and a presentation was made at ASGSR. • Tests were performed to evaluate the ability of BRE to emulate condensed fuels using gaseous fuels in normal gravity and in microgravity. • Tests were performed to calibrate the heat flux gages and paint and to better understand the effects on their output of the cooling water conditions. • Analytical models of the flames and burners were further developed. NOTE: This project is continued as a new grant with the same Principal Investigator--"Experimental Investigation of Emulated Burning Rate at Various Gravity Levels--NNX15AD06A." See that task for subsequent reporting. (Ed., 10/1/2015).
Bibliography Type:	Description: (Last Updated: 10/08/2025)
Abstracts for Journals and Proceedings	Zhang Y, Kim M, Sunderland PB, Quintiere JG, de Ris, Stocker DP. "A Burning Rate Emulator for Study in Microgravity." Presented at the 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. https://asgsr.org/index.php/presentation-abstracts-10-26-2014 ; accessed 10/6/15. , Oct-2014
Abstracts for Journals and Proceedings	Zhang Y, Kim M, Sunderland PB, Quintiere JG, de Ris, Stocker DP, Takahashi F, Ferkul PV. "A Burning Rate Emulator for Microgravity Studies." Poster, 35th International Symposium on Combustion, San Francisco, CA, August 3-8, 2014. 35th International Symposium on Combustion, San Francisco, CA, August 3-8, 2014. , Aug-2014
Articles in Peer-reviewed Journals	Zhang Y, Bustamante MJ, Gollner MJ, Sunderland PB, Quintiere JG. "Burning on flat wicks at various orientations." Journal of Fire Sciences. 2014 Jan;32(1):52-71. http://dx.doi.org/10.1177/0734904113495650 , Jan-2014
Dissertations and Theses	Kim M. "Measurement of Heat Flux in Burning Rate Emulator." M.S. Thesis, Dept. of Fire Protection Engineering, University of Maryland, expected Dec. 2014. , Dec-2014
Papers from Meeting Proceedings	Zhang Y, Kim M, Sunderland PB, Quintiere JG, de Ris J. "A Burner to Emulate Condensed Phase Fuels." Ninth Mediterranean Combustion Symposium, Rhodes, Greece, June 7-11, 2015. Ninth Mediterranean Combustion Symposium, Rhodes, Greece, June 7-11, 2015. 12 p. , Jun-2015